Malek Dinari

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Summary

Computer science engineer (graduated June 2025) with 13+ months of research and hands-on experience in Computer Vision, Machine Learning, and AI — including a research internship at CRISTAL Laboratory in collaboration with cAire, and ongoing work as a researcher in rPPG and contactless vital signs estimation. Currently building LearnLoop, a full-stack AI-powered study platform using local LLMs (FastAPI, Next.js, Ollama). Seeking an AI/ML Engineer or MLOps/LLMOps role in industry to apply my expertise in deep learning, computer vision, and end-to-end ML system design to production-grade solutions.

Education

Bachelor's Degree in Computer Science - ENSI SEPTEMBER 2021-June 2025

Status : Graduated as Computer Science engineer awaiting diploma issuance

Related Coursework: Object-oriented programming, Data Structures, Databases, Statistics, Computer Networks, Distributed Systems, Applied Mathematics, Computational Complexity, Quantization, Image Processing, Artificial Intelligence.

preparatory institute of engineering studies of Nabeul (IPEIN) 2019-2021

Mathematics Physics (MP): ranked 193/1779

Baccalaureate Mathematics 2018/2019

Graduated with 14.08/20

Projects

Full-Stack: LearnLoop - AI-Powered Adaptive Study Platform (WIP) GitHub Link

Personal project (February 2026 - Present):

- Building an intelligent study platform that generates quizzes from uploaded documents or free-text topics using a locally-running LLM (Ollama + Qwen 3.5 9B).
- Implemented full-stack architecture: FastAPI backend with document processing pipeline (PDF/TXT extraction, chunking, embedding), and Next.js 14 + TypeScript frontend with responsive quiz UI and real-time feedback.
- Core features operational: document upload with text extraction, multi-question quiz generation (MCQ, short-answer, true/false mix), instant LLM-graded feedback with explanations, and Socratic coaching chat.
- Achieved performance benchmarks: 30-120s for quiz generation, instant MCQ grading, 15-60s for short-answer grading. Identified bottlenecks (model thinking mode, 16K context window) and planned optimizations targeting 2-3x faster inference.
- High-priority roadmap: SSE streaming for real-time quiz question display (eliminates perceived wait), faster LLM model switching (Qwen 3:4B target), PostgreSQL + Redis for persistence and caching, Docker Compose containerization.
- Tech: Python (FastAPI, Pydantic), TypeScript, React, Next.js 14, Tailwind CSS, Ollama local LLM, pytest unit tests.

R&D: rPPG-Based Blood Pressure Estimation System (Ongoing) GitHub Link

Joint research & development project — CRISTAL Laboratory (ENSI, University of Manouba) & cAIre Healthcare AI (Industry Sponsor) *Feb 2025 – Present*

- Designed and developed a contactless blood pressure monitoring system leveraging remote photoplethysmography (rPPG) to extract physiological signals from standard video captures, enabling non-invasive estimation of systolic and diastolic blood pressure.
- Implemented a multi region-of-interest (ROI) anatomical site-based rPPG signal extraction pipeline: ROI detection, spatial patch decomposition, per-patch pulse signal recovery, and multi-region signal aggregation for robust cardiovascular feature estimation.
- Built an end-to-end self-supervised deep learning pipeline around TS-CAN (Temporal Shift Convolutional Attention Network), a multi-task attention-guided architecture that jointly optimizes rPPG signal recovery and an auxiliary attention concentration task via ridge regression, improving signal quality and reducing noise impact on the primary estimation task.
- Developed a real-time GUI for live blood pressure monitoring, visualization of extracted rPPG waveforms, and side-by-side comparison with contact-based reference measurements.
- Conducted extensive literature review and comparative analysis of state-of-the-art rPPG methods (deep learning and signal processing approaches), contributing structured technical summaries and methodological insights to ongoing research directions in contactless vital signs monitoring.
- Achieved promising experimental accuracy in BP estimation relative to traditional contact-based devices, advancing the state of the art in remote health monitoring.
- Tech: Python, PyTorch, OpenCV, NumPy/SciPy, signal processing (FFT, bandpass filtering, wavelet transforms), Docker, Matplotlib, Tkinter, conda, Ubuntu (WSL2)
- *Note: Closed-source project — code and proprietary datasets are confidential under collaboration with cAIre.*

Research Experience

Summer Research Internship - CRISTAL Laboratory SUMMER 2025 (2 months)

A Statistical Learning Approach to the Refinement of Geodesic Representations for 3D Face Analysis

- Conducted a 2-month research internship at CRISTAL Laboratory focused on 3D face analysis and geodesic shape representations.
- Studied and refined geodesic descriptors for curved surface analysis, with an emphasis on statistical learning methods for representation improvement and robustness.
- Worked on mesh / surface processing, geometric feature extraction, and evaluation of shape descriptors for face analysis tasks.
- Produced a written research report and consolidated the internship findings into a technical study on geometric representation for 3D face parametrization.

PhD Candidate / Researcher - CRISTAL Laboratory SEPTEMBER 2025-MARCH 2026

Reviewing and contributing to the state-of-the-art in remote Photoplethysmography (rPPG), heart rate, and blood pressure estimation from video data

- Engaged in a comprehensive review of current deep learning and signal processing approaches for camera-based rPPG extraction and vital sign estimation.
- Contributed to experimental analysis and comparative studies of state-of-the-art models in rPPG, focusing on performance, robustness, and reproducibility.
- Produced technical notes, structured summaries, and methodological insights to inform ongoing research directions in contactless vital signs monitoring.

Publications & Conferences

Poster — RFMI 2026 (IXème Atelier RFMI) January 8–9, 2026

Apprentissage machine pour l'estimation des paramètres vitaux

Malek Dinari, Enjie Ghorbel, Faouzi Ghorbel — CRISTAL Laboratory GRIFT Pôle, ENSI, Université de Manouba, in collaboration with cAIre.

- Presented a scientific poster at the IXème Atelier RFMI (Representations, Analysis and Recognition of Shape and Motion from Imaging Data), bringing together researchers in computer vision, machine learning, and medical imaging.
- Covered methodological approaches to improve the robustness of remote photoplethysmography (rPPG) pipelines for physiological parameter estimation (heart rate, blood pressure).
- Topics at the workshop included Riemannian and geometric ML, invariant theory, medical imaging, texture and colour analysis, and morphing attack detection.

Projects

Machine Learning/DL: Bone Marrow Smear Segmentation - ENSI GitHub Link

4-month design and development project:

- Built an AI pipeline to classify and segment bone marrow cells, providing biologists with a diagnostic tool to aid in disease monitoring.
- Fine-tuned MobileNet for initial image segmentation, followed by inference with YOLOv8 to detect multiple cell types within smears, enabling comprehensive cell identification in a single workflow.
- Deployed an interactive frontend using Gradio in a Colab Notebook for accessibility and ease of use in research or clinical settings.
- This tool is tailored for biologists to facilitate bone marrow analysis and improve diagnostic accuracy through automated image analysis.

Mobile Development: WeeFizz Mobile Size Recommendation App - Welyne GitHub Link

3-month summer internship project:

- Developed the mobile app for WeeFizz, a size recommendation tool, as part of a project with Welyne, an IT services startup. The app was built using React Native (community CLI, bare workflow) and integrates seamlessly with the existing WebApp.
- Implemented core functionalities, including QR code scanning (or manual URL entry), user parameter collection (e.g., height, body type), and photo capture for personalized size suggestions.
- Designed to support Prestashop users, the app enhances the shopping experience by offering precise clothing size recommendations based on individual body measurements.
- Gained mobile development experience in the context of practical e-commerce solutions while developing a deeper interest in roles focused on ML and DevOps.

ML/NLP & AI: Mini-GPT Model - Self-Directed GitHub Link

Mini-project (in progress):

- Implemented a foundational GPT-like model from scratch using Shakespeare's complete works as training data, with a focus on understanding core concepts in generative pre-trained transformers and attention mechanisms.
- Designed and trained an initial Bigram Language Model as a baseline for autoregressive text generation, while exploring scalable architectures that enhance linguistic coherence and depth.
- Expanding the model to incorporate trigram and n-gram components, with a future goal of integrating a multilayer perceptron (MLP) for increased capacity and scalability in handling more complex language structures.
- This project aims to reinforce knowledge in natural language processing (NLP), particularly in attention-based architectures, while providing a stepping stone towards developing larger models or advanced text generation pipelines.

Projects

ML: Salary Calculator Interface Github Link

- Trained and deployed an AI model via Streamlit to calculate salaries based on the 2020 Stack Overflow Developer Survey.

Computer Vision: 3D Facial Expression Recognition - ENSI GitHub Link

Part of a university research project supervised by Prof. Faouzi GHORBEL & Dr-Ing. Majdi JRIBI) :

- Developing algorithms for facial expression recognition using 3D representations (curved 2D surfaces, point clouds, or meshes).
- Implementing geodesic potential descriptors for level curve extraction and geodesic distance computation (Dijkstra's algorithm).
- Exploring advanced Computer Vision techniques such as SIFT detectors and Iterative Closest Point (ICP) for mesh matching.
- Aimed at creating robust 3D or 4D (3D+t) models for improved biometric and expression classification.
- Tools & Skills: Python, Matplotlib, Plotly, PCA, Linear Algebra, 3D Visualization.

Selected Computer Vision & ML Notebooks GitHub Link

Ongoing personal project:

- Created a structured set of Jupyter notebooks covering fundamental techniques and algorithms in computer vision and machine learning, organized into four primary modules:
  - **Pattern Recognition:** Implementation of essential techniques such as Hu's 7 moments, Ghorbel complex moments invariance, and Jan Flusser's rotation invariants.
  - **Mathematical Morphology:** Exploration of morphological operations, including geodesic dilation reconstruction, erosion-dilation, and advanced Laplacian and top-hat transformations.
  - **Image Processing and Restoration:** Covering key concepts like DFT/IDFT, Fast Fourier Transform (FFT), Gaussian and motion blurring, kernel-based filtering, and Wiener filters, with additional introductory work in deep learning and optimization.
  - **3D Mesh Processing & Face Expression Analysis:** Focused on facial expression recognition and 3D mesh representation, including level curve analysis using geodesic distances and methods for representing curves on facial meshes.
- Aims to provide a comprehensive reference for foundational and advanced topics in computer vision and image processing, with ongoing updates and new notebooks planned or to be added.

Full-Stack Application: Provider Management System GitHub Link

- Developed a complete CRUD application using Spring Boot for the backend and Angular for the frontend, hosted on PHPMyAdmin for database management.
- Implemented user-friendly features, such as a provider list display and a form for adding new providers, showcasing strong full-stack expertise.
- Gained experience in Spring Suite 4, REST APIs, and Angular component-based design for dynamic web single-page applications.

NLP/C Programming: Semantic and syntactic analyzer Github Link

- Created an app that performs semantic and syntactic analysis for sentences and expressions.

Certificates & Licenses

- **Build Multimodal Generative AI Applications**, IBM — Issued Mar 2026
- **Advanced RAG with Vector Databases and Retrievers**, IBM — Issued Mar 2026

- **Vector Databases for RAG: An Introduction**, IBM — Issued Feb 2026
- **Build RAG Applications: Get Started**, IBM — Issued Feb 2026
- **Develop Generative AI Applications: Get Started**, IBM — Issued Feb 2026
- **Intermediate Docker**, DataCamp — Issued Jan 2026
- **Advanced Git**, DataCamp — Issued Jan 2026
- **Transformer Models with PyTorch**, DataCamp — Issued Nov 2025

Relevant Skills

Programming languages & Frameworks	C/C++/Java-Intermediate, JavaScript-Intermediate, TypeScript-Beginner,SQL-Intermediate, Python-Intermediate, PyTorch/Tensorflow/Keras-Intermediate, React Native-Intermediate, NodeJS-Beginner, Flutter-Beginner, Angular-Intermediate, Spring Boot-Intermediate
Technologies & Tools	Kaggle/Colab/Jupyter Notebook, DevOps tools(Git/GitHub, Docker), Gradio, Postman (REST APIs testing), Streamlit
Language Skills	English-Fluent, Arabic-Native, French-Proficient, Spanish-A1:Beginner, German-A1:Beginner

Interests

Machine Learning	Reading & Researching	Music
Mathematics	Data Science	Chess and Strategy Games

Last updated: April 10, 2026